Algorithms and Data Structures

REPORT LAB 6

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TASK1

```
                > & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win3
r2.ydr' '--stderr=Microsoft-MIEngine-Error-pg5m1bup.i0c' '--pid=Microsoft-MIEngine-Pid-
--- Task 1-1: Inserting elements (15, 10, 30, 40, 5) ---

Inserting 15:
The value of A[1] before increase is: -2147483648
The value of A[1] after increase is: 15
Current heap: 15

Inserting 10:
The value of A[2] before increase is: -2147483648
The value of A[2] after increase is: 10
Current heap: 15 10

Inserting 30:
The value of A[3] before increase is: -2147483648
The value of A[3] after increase is: 30
We need to swap the values of 30, 15
Current heap: 30 10 15

Inserting 40:
The value of A[4] before increase is: -2147483648
The value of A[4] after increase is: 40
We need to swap the values of 40, 10
We need to swap the values of 40, 30
Current heap: 40 30 15 10

Inserting 5:
The value of A[5] before increase is: -2147483648
The value of A[5] after increase is: 5
Current heap: 40 30 15 10 5

--- Task 1-2: Finding Maximum ---
Maximum element is: 40

--- Task 1-3: Extracting Maximum ---
Extracted: 40
Heap after extraction: 30 10 15 5

--- Task 1-4: Increasing index 3 to 35 ---
The value of A[3] before increase is: 15
The value of A[3] after increase is: 35
We need to swap the values of 35, 30
Heap after increase-key: 35 10 30 5

--- Task 1-5: Decreasing index 4 to 2 ---
Heap after decrease-key: 35 10 30 2
PS C:\Users\ILGIN>
```

TASK1

```
55.upk' '--stderr=Microsoft-MIEngine-Error-u3exprov.apt' '--pid=Microsoft-MIEngine-Pid-4yi4k225.
--- Task 2-1: Counting Sort ---
Original array A: 2 5 3 0 2 3 0 3

Step 2: array C after counting each element's frequency is : 2 0 2 3 0 1
Step 3: compute the cumulative counts of array C step by step
2 2 2 3 0 1
2 2 4 3 0 1
2 2 4 7 0 1
2 2 4 7 7 1
2 2 4 7 7 8

-------------Step 4--------------
array A is : 2 5 3 0 2 3 0 3
array C is : 2 2 4 7 7 8
array B is : 0 0 -1014032064 32758 0 0 3 0

-------------Step 4--------------
array A is : 2 5 3 0 2 3 0 3
array C is : 2 2 4 6 7 8
array B is : 0 0 -1014032064 32758 0 0 3 0

-------------Step 4--------------
array A is : 2 5 3 0 2 3 0 3
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array A is : 2 5 3 0 2 3 0 3
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-------------Step 4--------------
array A is : 2 5 3 0 2 3 0 3
array C is : 0 2 3 4 7 8
array B is : 0 0 -1014032064 2 3 3 3 5

-------------Step 4--------------
array A is : 2 5 3 0 2 3 0 3
array C is : 0 2 3 4 7 7
array B is : 0 0 2 2 3 3 3 5

Final sorted array B: 0 0 2 2 3 3 3 5
PS C:\Users\ILGIN>
```